- Measure σ.

The repeated experiment spits out a random series of plus-ones and minus-ones. Determinism has broken down, but in a particular way. If we do many repetitions, we will find that the numbers of $\sigma = +1$ events and $\sigma = -1$ events are statistically equal. In other words, the average value of σ is zero. Instead of the classical result—namely, that the component of σ along the x axis is zero—we find that the *average of these repeated measurements* is zero.

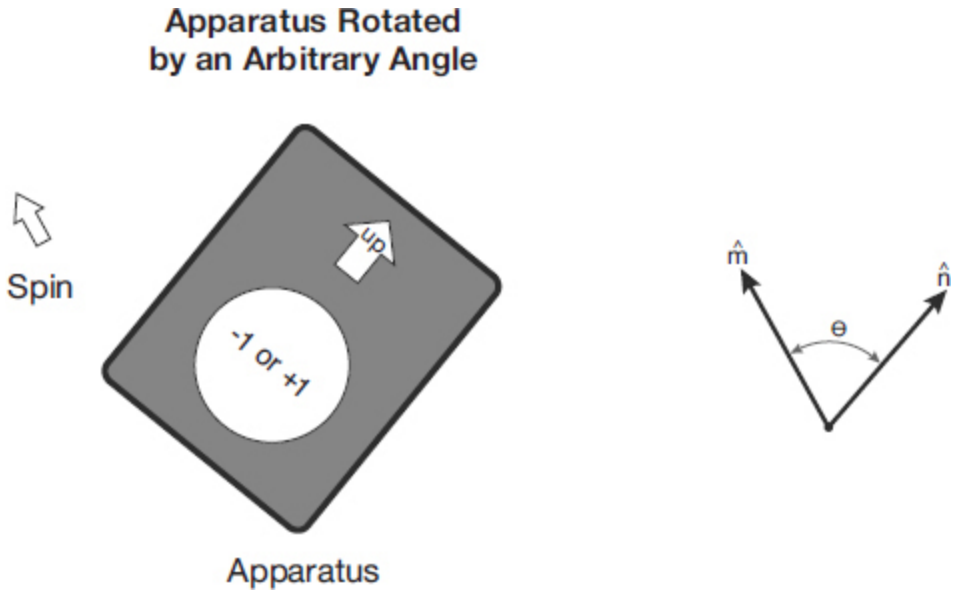


Figure 1.4: The apparatus rotated by an arbitrary angle within the x–z plane. Average measurement result is $\hat{n} \cdot \hat{m}$.

Now let's do the whole thing over again, but instead of rotating $\mathcal{A}$ to lie on the x axis, rotate it to an arbitrary direction along the unit vector[2] $\hat{n}$. Classically, if σ were a vector, we would expect the result of the experiment to be the component of σ along the $\hat{n}$ axis. If $\hat{n}$ lies at an angle θ with respect to z, the classical answer would be $\sigma = \cos\theta$. But as you might guess, each time we do the experiment we get $\sigma = +1$ or $\sigma = -1$. However, the result is statistically biased so that the average value is $\cos\theta$.

The situation is of course more general. We did not have to start with $\mathcal{A}$ oriented along z. Pick any direction $\hat{m}$ and start with the up-arrow pointing